

# **JNTU Kakinada**

## **REGULAR EXAMINATION ? 2023**

Probability and Statistics for Data Science ? Regular Examination, 2023

Subject Code: DS301 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Descriptive Statistics)**

1. Solve a problem related to Descriptive Statistics. Show all the steps clearly. [12 Marks]
2. Derive the expression related to Descriptive Statistics and explain each term. [12 Marks]
3. Describe the working principle of Descriptive Statistics with necessary equations. [12 Marks]

### **SECTION B (Unit 2: Probability Distributions)**

4. Solve a problem related to Probability Distributions. Show all the steps clearly. [12 Marks]
5. Solve a problem related to Probability Distributions. Show all the steps clearly. [12 Marks]
6. Derive the expression related to Probability Distributions and explain each term. [12 Marks]

### **SECTION C (Unit 3: Statistical Inference)**

7. Define Statistical Inference. Explain its significance in Probability and Statistics for Data Science with a suitable example. [12 Marks]
8. Describe the working principle of Statistical Inference with necessary equations. [8 Marks]
9. Discuss the engineering applications of Statistical Inference. [12 Marks]

### **SECTION D (Unit 4: Correlation and Regression)**

10. List the advantages and limitations of Correlation and Regression. Explain any two in detail. [12 Marks]
11. How is Correlation and Regression implemented? Give an algorithm / procedure and explain. [12 Marks]
12. Compare and contrast the key concepts involved in Correlation and Regression. [12 Marks]

### **SECTION E (Unit 5: Hypothesis Testing)**

13. Explain Hypothesis Testing with a neat diagram/sketch and a suitable example. [12 Marks]

**Marks]**

14. Define Hypothesis Testing. Explain its significance in Probability and Statistics for Data Science with a suitable example. [12 Marks]

15. Explain Hypothesis Testing with a neat diagram/sketch and a suitable example. [12 Marks]

\*\*\* End of Question Paper \*\*\*